Attribute	Java	C	C++	Python	JavaScript
Language/ Scripting	Programming	Programming	Programming	Scripting	Scripting
Open source/ Commercial	Partly Open Source	Open Source	Open Source	Open Source	Partly Open Source
Compiler/ Interpreter	Compiler	Compiler	Compiler	Interpreter	Interpreter
Object Oriented-Programming Support (Yes/No)	Yes	No	Yes	Yes	Yes
Developer Organization	Sun Microsystems	AT&T Bell Laboratories	AT&T Bell Laboratories	Python Software Foundation	Netscape
Developer (Person)	James Gosling	Dennis Ritchie	Bjarne Stroustrup	Guido Van Rossum	Brendan Eich
Current Major Version (2026)	Java26	C23	C++26	Python 3.14	ECMA Script 2025 (ES16)
Primary Purpose	To let Programmers, Write once, Run anywhere (WORA)	To write and implement the UNIX Operating System	To add OOP features to the pre-existing C language	To act as a Versatile, general - purpose programming language	To add interactivity and dynamic behavior to websites
Common Applications	Android mobile apps, Enterprise and Financial systems, Cloud	Developing OS, DBMS, embedded systems, compilers	Low-level hardware control, high execution speed, efficient memory management	Data Science, Machine Learning, Artificial Intelligence, Web development	Web and mobile development, VR, AI
Database Support	Yes	Yes	Yes	Yes	Yes
Memory Management	Garbage Collection	Manual	Manual	Garbage Collection	Garbage Collection
Security Features	High	Low	Low	High	Medium
Performance	Fast/Moderate	Maximum Speed	Maximum Speed	Slower than compiled languages	Moderate

Platform Independence (Yes/No)	Yes	No	No	Yes	Yes
Other Important Features	Strong JVM ecosystem, Multithreading support	Small language size, direct hardware access	Templates, STL (Standard Template Library)	Extensive Libraries, Rapid prototyping	Event-driven, asynchronous programming, runs is browsers
Ease of Learning	Moderate	Hard	Very Hard	Very Easy	Moderate
Popular IDEs	IntelliJ IDEA	Visual Studio	CLion	PyCharm	WebStorm
Compilation Output	Bytecode	Machine Code	Machine Code	Bytecode	Just-In-Time machine code
Advantages	High security	Exceptional execution speed	Multi-paradigm support	Simplicity in learning	Interactive web pages, large ecosystem, cross-platform support
Limitations	Higher memory usage, slower startup time	No built-in OOP, Manual memory management	Complex syntax, Difficult debugging	Lower execution speed, Higher memory consumption	Browser compatibility issues, Dynamic typing can cause runtime errors

Requirement	Most Suitable Language	Reason
System Programming	C	Direct hardware access, high performance, widely used for operating systems.
Enterprise Applications	Java	Secure, scalable, platform-independent, and widely used in enterprise software.
Web Development	JavaScript	Standard language for interactive web applications and supported by all modern browsers
Artificial Intelligence	Python	Rich ecosystem of AI and machine learning libraries

		like TensorFlow, PyTorch, and scikit-learn.
Mobile Application Development	Java	Traditionally used for Android development and still widely supported.

Conclusion:

The comparison of Java, C, C++, Python, and JavaScript shows that each programming language has unique strengths and is designed for different purposes. C and C++ are ideal for system-level programming because of their high performance and direct memory control. Java is widely used for enterprise and Android application development due to its platform independence, security, and reliability. Python is the preferred language for artificial intelligence, data science, and rapid application development because of its simple syntax and extensive libraries. JavaScript is the backbone of modern web development, enabling interactive and dynamic websites. Overall, no single language is best for every task; selecting the appropriate language depends on the project's requirements, performance needs, and development goals.

Installation of Java and Execution of a Simple Hello World Program (Windows) in Oracle JDK

Software Required

- Windows Operating System
- Internet Connection
- Java Development Kit (JDK)

Step 1: Download Java JDK

1. Open a web browser.
2. Visit the official Oracle JDK website: Oracle JDK Downloads
3. Download the latest JDK for Windows (e.g., JDK 25).

Step 2: Install JDK

1. Double-click the downloaded .exe file.
2. Click **Next** and follow the installation wizard.
3. Complete the installation.

Step 3: Verify Java Installation

1. Open **Command Prompt**.
2. Type: java --version

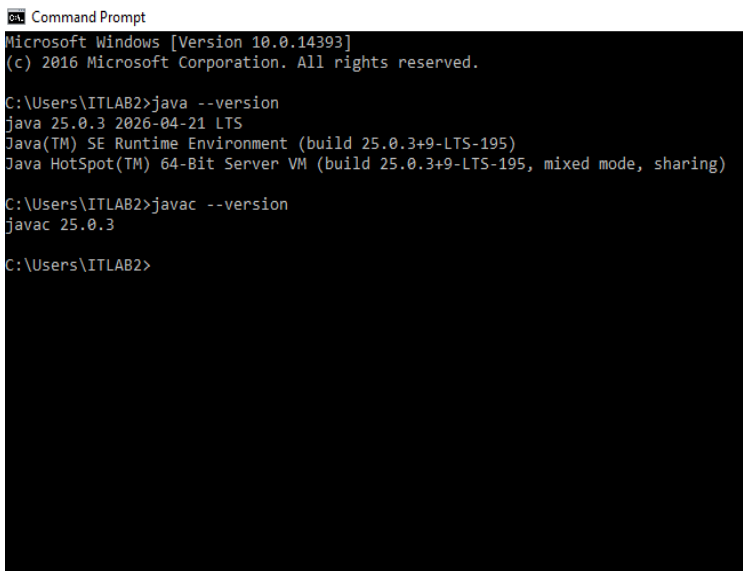
Sample Output:

java 25.0.3

3. Check the Java Compiler: javac --version

Sample Output:

javac 25.0.3



Configuring Java on Windows (Setting JAVA_HOME and PATH)

Step 1: Find Java Installation Path

After installing Java, locate the JDK folder.

Example:

C:\Program Files\Java\jdk-25.0.3

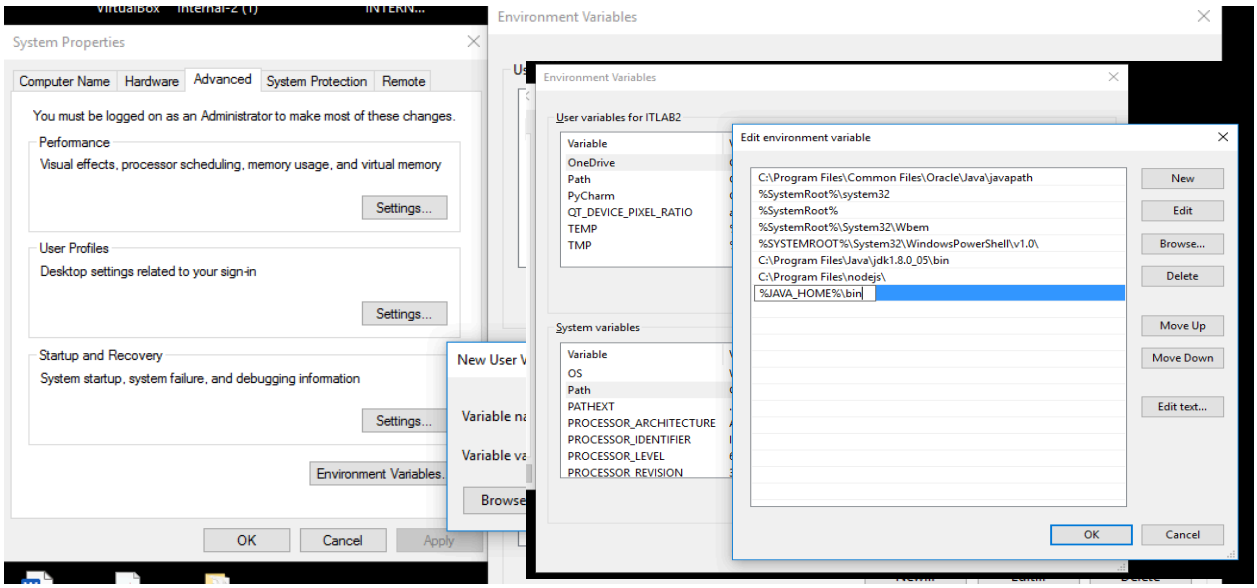
Step 2: Set JAVA_HOME

1. Press **Windows + R** → Type sysdm.cpl → Press Enter.
2. Select **Advanced** tab.

3. Click **Environment Variables**.
4. Under **System Variables**, click **New**.

Enter:

Variable Name : JAVA_HOME
Variable Value: C:\Program Files\Java\jdk-25.0.3



5. Click **OK**.

Step 3: Update PATH Variable

1. Under **System Variables**, select **Path**.
2. Click **Edit**.
3. Click **New**.
4. Add:

%JAVA_HOME%\bin

5. Click **OK → OK → OK**.

Step 4: Verify Configuration

Open a new Command Prompt and execute:

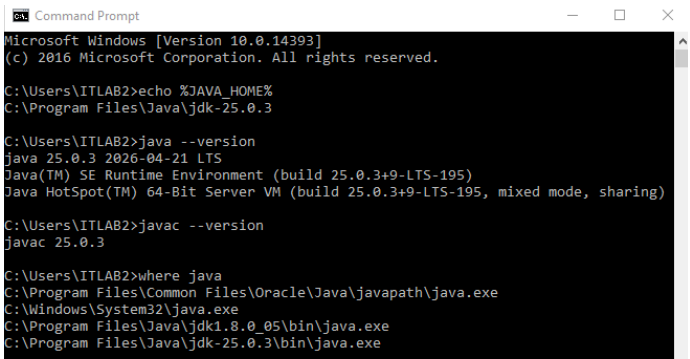
echo %JAVA_HOME%

Output:

C:\Program Files\Java\jdk-25.0.3

Check Java version:

java --version



Check Java compiler:

javac --version

Sample Output:

java 25.0.3
javac 25.0.3

Execution of ‘Hello’ program in Java:

1. Create a Java file:

Create a file named Hello.java:

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello");
    }
}
```

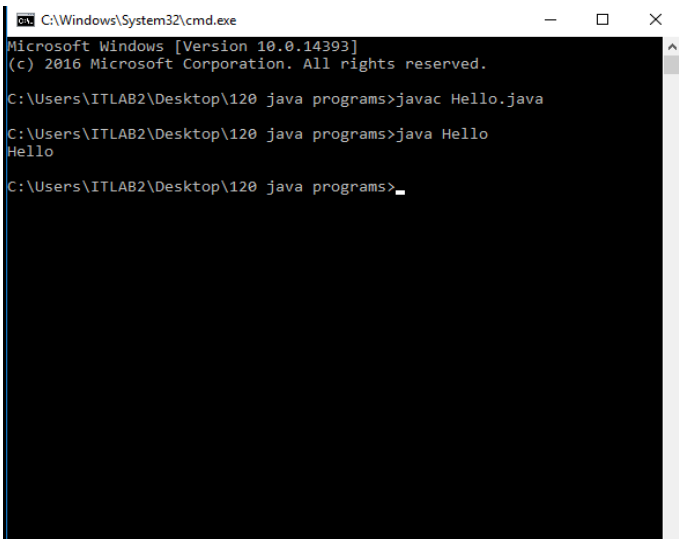
Store it in the format of .java

2. Open Command Prompt:

Type javac Hello.java

It would create a ‘.class’ file for the program.

3. Run the program:
Type java Hello
Sample Output:
Hello



Installation of Java and Execution of a Simple Hello World Program (Windows) in OpenJDK

Software Required:

- Windows Operating System
- Internet Connection
- OpenJDK
- Command Prompt

Step 1: Download OpenJDK

1. Open any web browser.
2. Visit the official **OpenJDK** website or **Eclipse Adoptium** website.
3. Download the latest stable **OpenJDK for Windows (64-bit)**.

Step 2: Install OpenJDK

1. Double-click the downloaded installer.
2. Follow the installation wizard.
3. Choose the installation location if required.
4. Click **Install** and wait until the installation is completed.
5. Click **Finish**.

Step 3: Configure Java Environment Variables

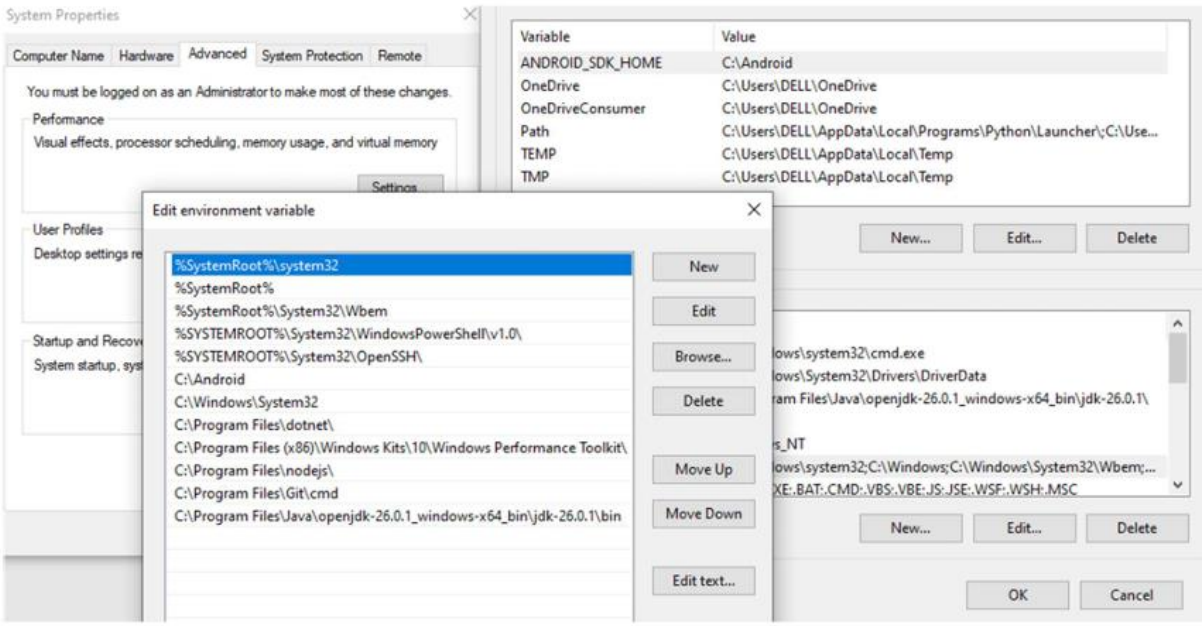
Set JAVA_HOME

1. Press **Windows + R**, type **sysdm.cpl**, and press **Enter**.
2. Open the **Advanced** tab.
3. Click **Environment Variables**.
4. Under **System Variables**, click **New**.
5. Enter the following:

Variable Name: JAVA_HOME

Variable Value: *(Example)*

C:\Program Files\Eclipse Adoptium\jdk-21



(Use the path where OpenJDK is installed on your system.)

Update the PATH Variable

1. Under **System Variables**, select **Path**.
2. Click **Edit**.
3. Click **New**.
4. Add:

%JAVA_HOME%\bin

5. Click **OK** to save all the changes.

Step 4: Verify the Installation

1. Open **Command Prompt**.
2. Type:

java --version

Sample Output

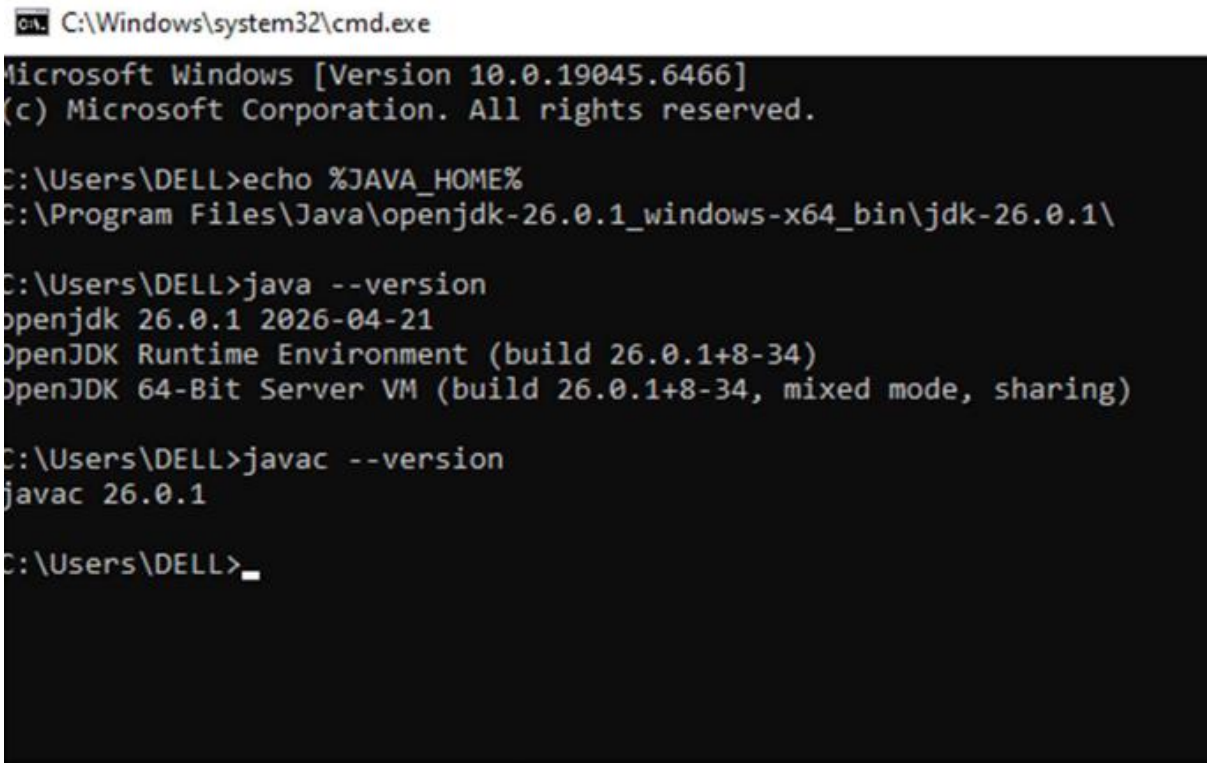
openjdk 21.0.11

3. Now check the Java compiler.

javac --version

Sample Output

javac 21.0.11



Execution of a Simple Hello World Program

Step 1: Create a Java Program

Create a file named **Hello.java** and type the following program:

```
public class Hello {
   public static void main(String[] args) {
      System.out.println("Hello World");
   }
}
```

Save the file with the **.java** extension.

Step 2: Compile the Program

Open Command Prompt and navigate to the folder where the file is saved.

Compile the program using:

javac Hello.java

A file named **Hello.class** will be generated after successful compilation.

Step 3: Run the Program

Execute the program using:

java Hello

Output

Hello World

